Assignment

An assignment for class **MET CS 575** at Boston University, instructed by **Dr. Nourai**. Student Zhang, Jin Yu submitted on **October 27, 2022**.

Step 1: Create Hello World Programs

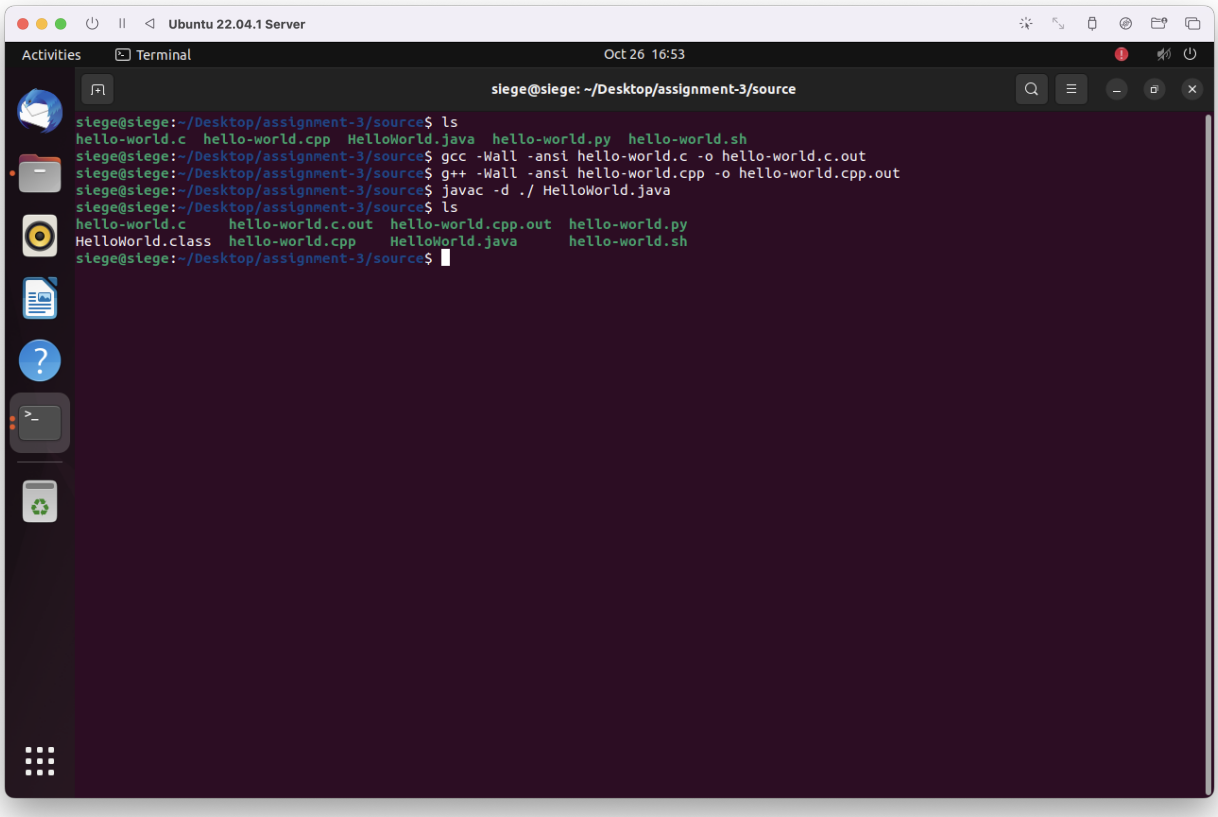
Input Filename	Command Used for Compiling	Output Filename
`hello-world.c`	`gcc -Wall -ansi hell-world.c -o hello-world.c.out`	`hello-world.c.out`
`hello-world.cpp`	`g++ -Wall -ansi hell-world.cpp -o hello-world.cpp.out`	`hello-world.cpp.out`
`hello-word.py`		
`hello-world.sh`		
`HelloWorld.java`	`javac -d ./ HelloWorld.java`	`HelloWorld.class`

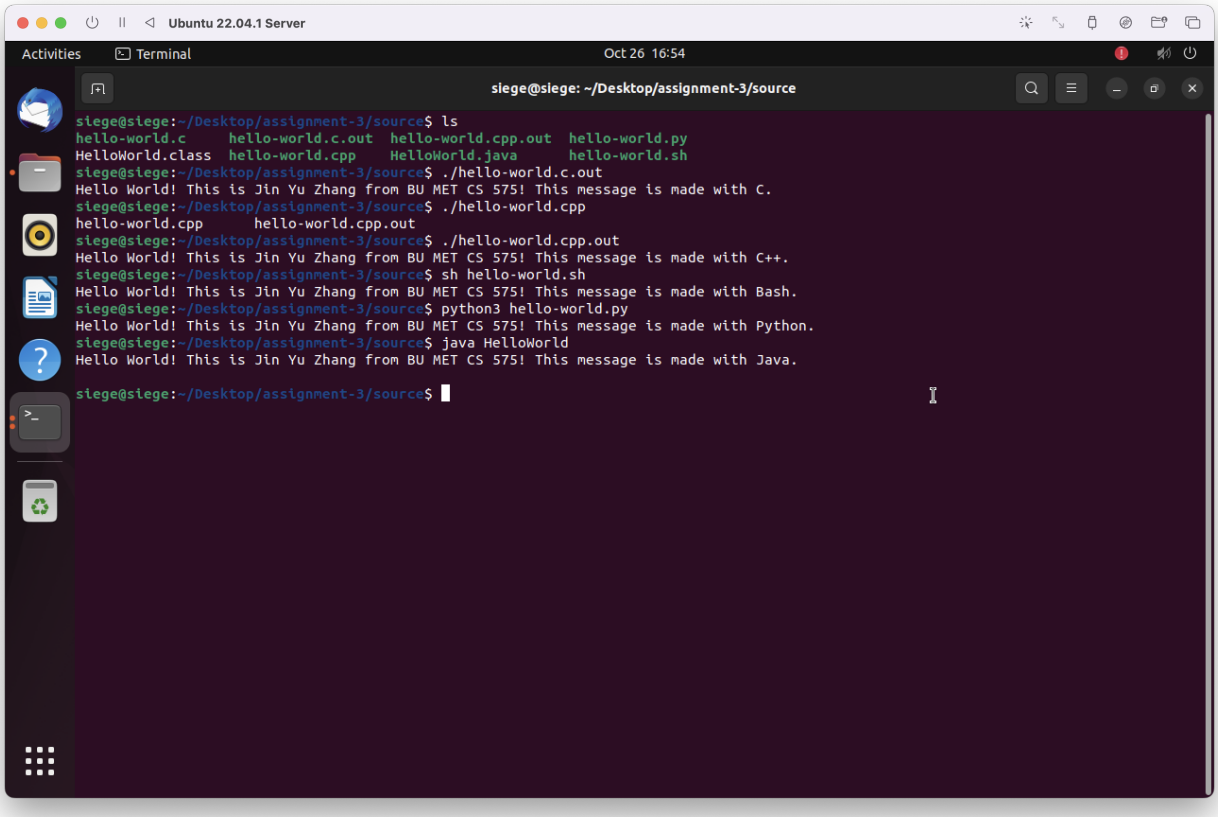
Java

I had no issues while using `gcc`, `g++`, and `python3`. However, I had to run a few steps to install Java Compiler. `jdk-8u351-linux-aarch64.tar.gz` has been downloaded and put in the shared folder:

```
cd <shared_folder>
cp jdk-8u351-linux-aarch64.tar.gz ./jdk-8
chmod 666 jdk-8

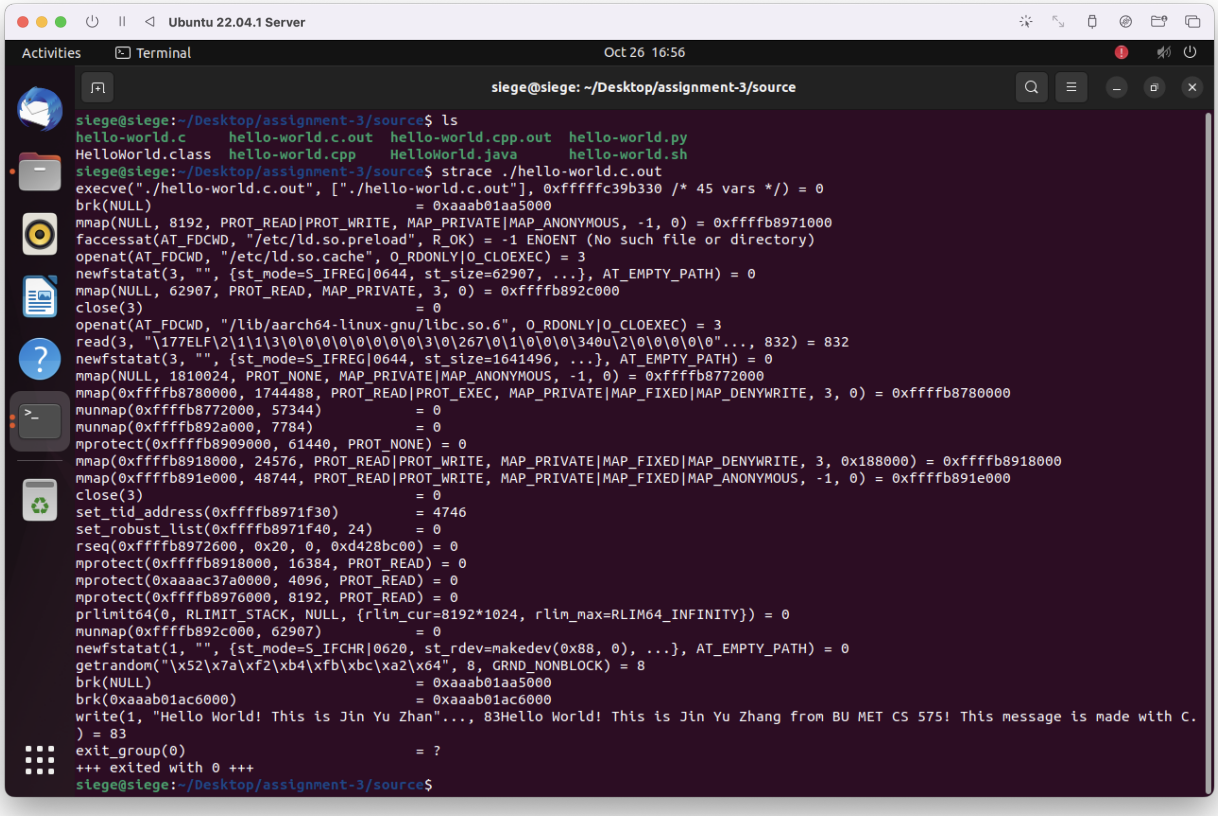
sudo mkdir /usr/local/java
cp jdk-8 /usr/local/java/
sudo tar zxvf jre-8
sudo rm jre-8
sudo update-alternatives --install "/usr/bin/java" "java" "/usr/local/java/jdk1.8.0_351/bin/java" 1
sudo update-alternatives --install "/usr/bin/javac" "javac" "/usr/local/java/jdk1.8.0_351/bin/javac" 1
```

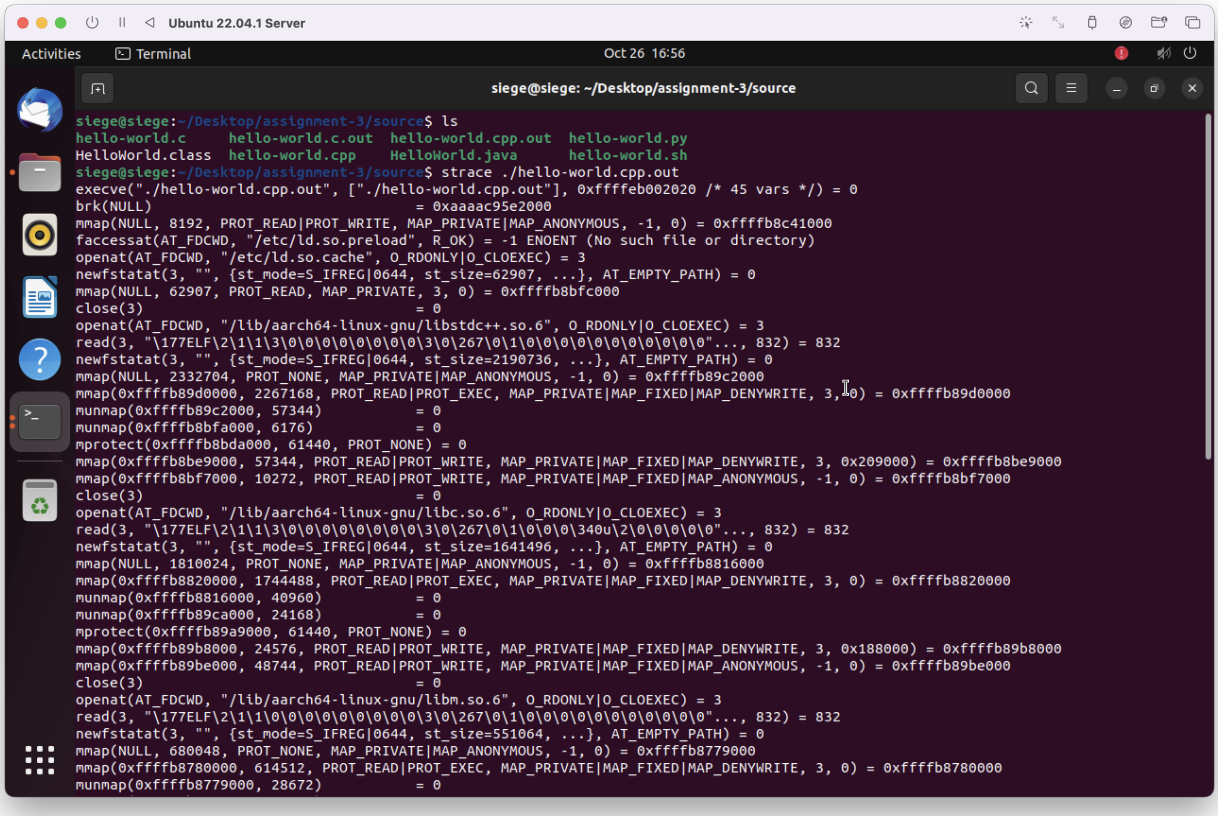


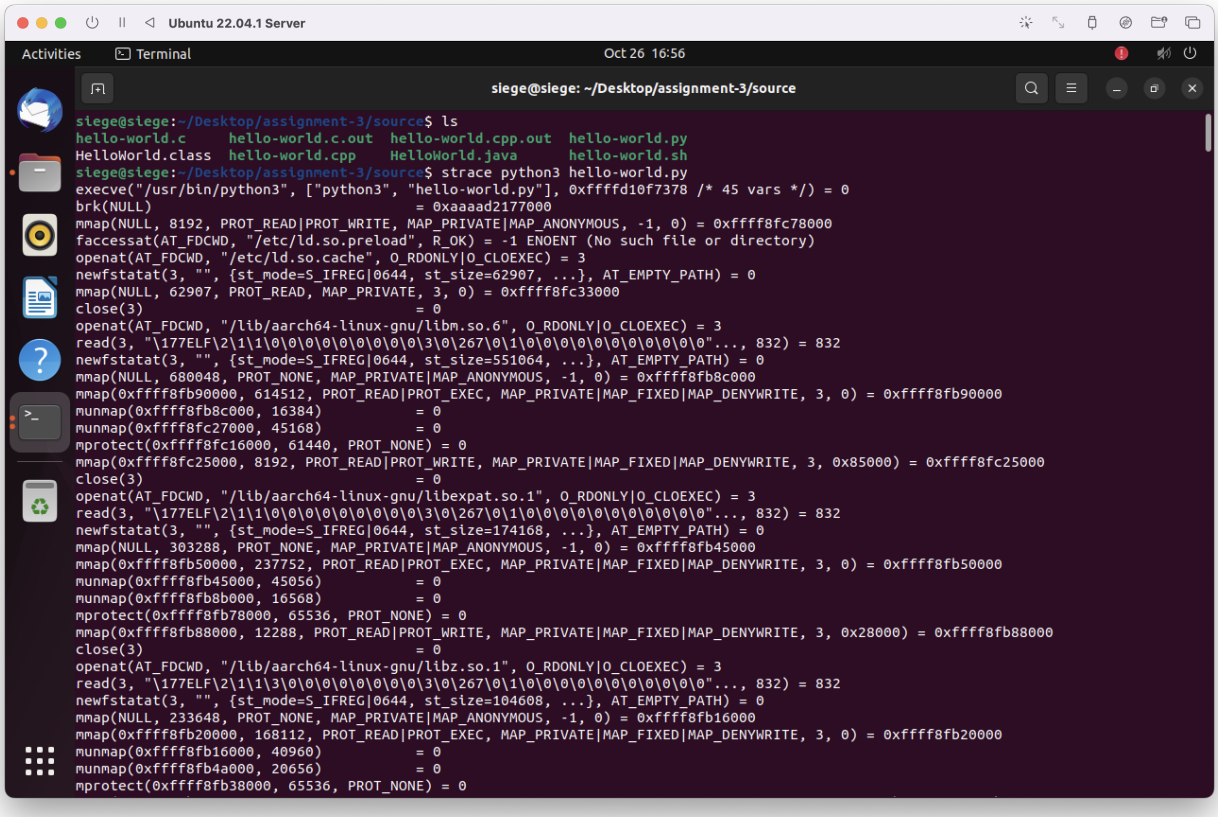

Ubuntu 22.04.1 Server
Activities
Terminal
Oct 26 16:54
siege@siege: ~/Desktop/assignment-3/source
siege@siege:~/Desktop/assignment-3/source$ ls
hello-world.c hello-world.c.out hello-world.cpp.out hello-world.py
HelloWorld.class hello-world.cpp HelloWorld.java hello-world.sh
siege@siege:~/Desktop/assignment-3/source$ ./hello-world.c.out
Hello World! This is Jin Yu Zhang from BU MET CS 575! This message is made with C.
siege@siege:~/Desktop/assignment-3/source$ ./hello-world.cpp
hello-world.cpp hello-world.cpp.out
siege@siege:~/Desktop/assignment-3/source$ ./hello-world.cpp.out
Hello World! This is Jin Yu Zhang from BU MET CS 575! This message is made with C++.
siege@siege:~/Desktop/assignment-3/source$ sh hello-world.sh
Hello World! This is Jin Yu Zhang from BU MET CS 575! This message is made with Bash.
siege@siege:~/Desktop/assignment-3/source$ python3 hello-world.py
Hello World! This is Jin Yu Zhang from BU MET CS 575! This message is made with Python.
siege@siege:~/Desktop/assignment-3/source$ java HelloWorld
Hello World! This is Jin Yu Zhang from BU MET CS 575! This message is made with Java.
siege@siege:~/Desktop/assignment-3/source$

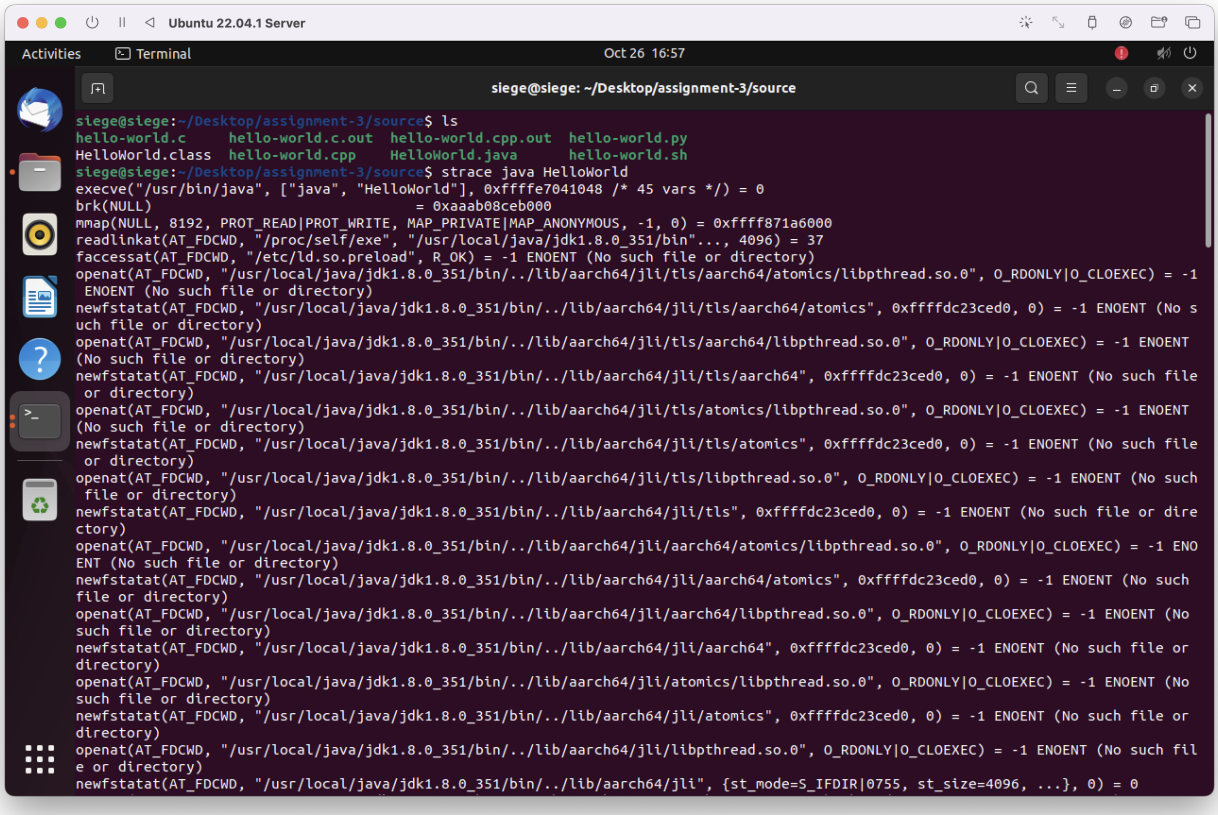
Step 2: Execute Hello World Programs

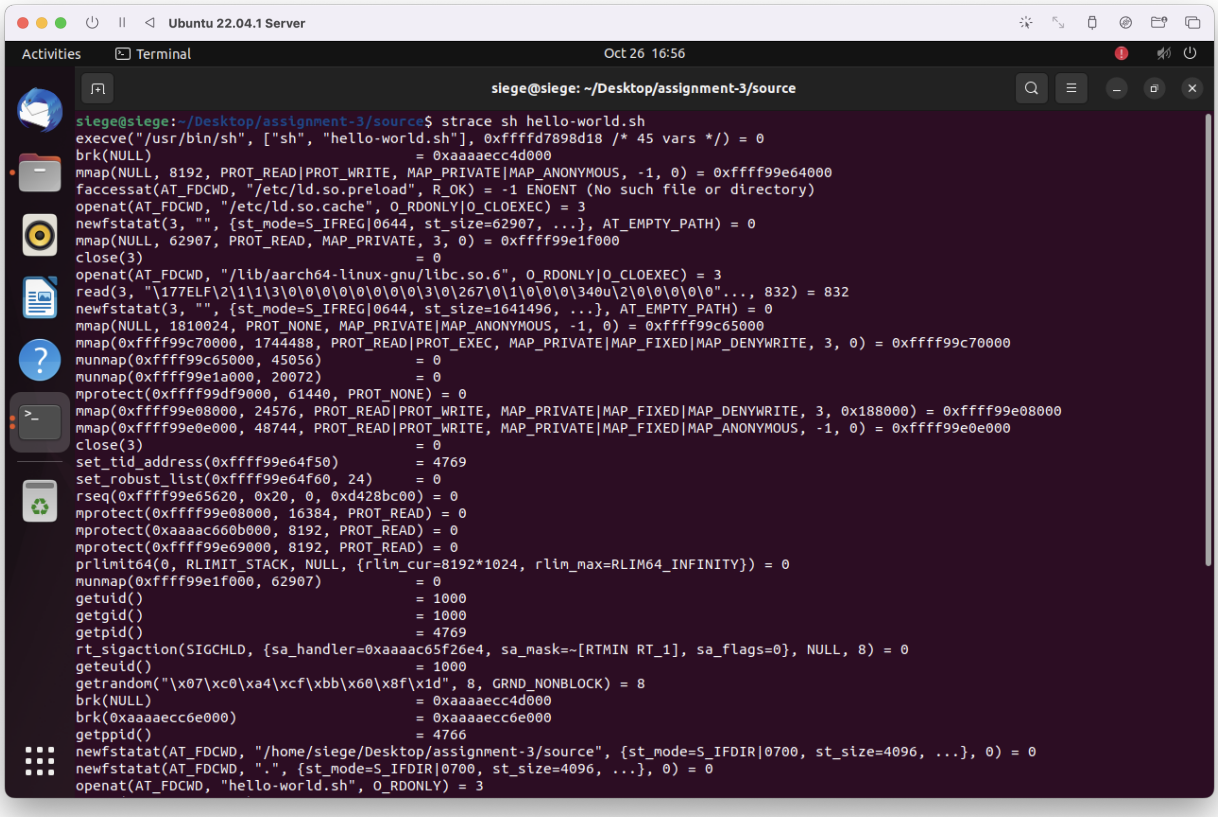
Execute each Hello World program with `strace`:






Ubuntu 22.04.1 Server
Activities
Terminal
Oct 26 16:56
siege@siege: ~/Desktop/assignment-3/source
siege@siege:~/Desktop/assignment-3/source$ ls
hello-world.c hello-world.c.out hello-world.cpp.out hello-world.py
HelloWorld.class hello-world.cpp HelloWorld.java hello-world.sh
siege@siege:~/Desktop/assignment-3/source$ strace python3 hello-world.py
execve("/usr/bin/python3", ["python3", "hello-world.py"], 0xffffd10f7378 /* 45 vars */) = 0
brk(NULL) = 0xaaaad2177000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xffff8fc78000
faccessat(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=62907, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 62907, PROT_READ, MAP_PRIVATE, 3, 0) = 0xffff8fc33000
close(3) = 0
openat(AT_FDCWD, "/lib/aarch64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=551064, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 680048, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xffff8fb8c000
mmap(0xffff8fb90000, 614512, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xffff8fb90000
munmap(0xffff8fb8c000, 16384) = 0
munmap(0xffff8fc27000, 45168) = 0
mprotect(0xffff8fc16000, 61440, PROT_NONE) = 0
mmap(0xffff8fc25000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x85000) = 0xffff8fc25000
close(3) = 0
openat(AT_FDCWD, "/lib/aarch64-linux-gnu/libexpat.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=174168, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 303288, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xffff8fb45000
mmap(0xffff8fb50000, 237752, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xffff8fb50000
munmap(0xffff8fb45000, 45056) = 0
munmap(0xffff8fb8b000, 16568) = 0
mprotect(0xffff8fb78000, 65536, PROT_NONE) = 0
mmap(0xffff8fb88000, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0xffff8fb88000
close(3) = 0
openat(AT_FDCWD, "/lib/aarch64-linux-gnu/libz.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=104608, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 233648, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xffff8fb16000
mmap(0xffff8fb20000, 168112, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xffff8fb20000
munmap(0xffff8fb16000, 40960) = 0
munmap(0xffff8fb4a000, 20656) = 0
mprotect(0xffff8fb38000, 65536, PROT_NONE) = 0


Ubuntu 22.04.1 Server
Activities
Terminal
Oct 26 16:57
siege@siege: ~/Desktop/assignment-3/source
siege@siege:~/Desktop/assignment-3/source$ ls
hello-world.c hello-world.c.out hello-world.cpp.out hello-world.py
HelloWorld.class hello-world.cpp HelloWorld.java hello-world.sh
siege@siege:~/Desktop/assignment-3/source$ strace java HelloWorld
execve("/usr/bin/java", ["java", "HelloWorld"], 0xffffe7041048 /* 45 vars */) = 0
brk(NULL) = 0xaaab08ceb000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xffff871a6000
readlinkat(AT_FDCWD, "/proc/self/exe", "/usr/local/java/jdk1.8.0_351/bin"..., 4096) = 37
faccessat(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/tls/aarch64/atomics/libpthread.so.0", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
newfstatat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/tls/aarch64/atomics", 0xffffdc23ced0, 0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/tls/aarch64/libpthread.so.0", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
newfstatat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/tls/aarch64", 0xffffdc23ced0, 0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/tls/atomics/libpthread.so.0", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
newfstatat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/tls/atomics", 0xffffdc23ced0, 0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/tls/libpthread.so.0", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
newfstatat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/tls", 0xffffdc23ced0, 0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/aarch64/atomics/libpthread.so.0", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
newfstatat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/aarch64/atomics", 0xffffdc23ced0, 0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/aarch64/libpthread.so.0", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
newfstatat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/aarch64", 0xffffdc23ced0, 0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/atomics/libpthread.so.0", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
newfstatat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/atomics", 0xffffdc23ced0, 0) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli/libpthread.so.0", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
newfstatat(AT_FDCWD, "/usr/local/java/jdk1.8.0_351/bin/../lib/aarch64/jli", {st_mode=S_IFDIR|0755, st_size=4096, ...}, 0) = 0


Ubuntu 22.04.1 Server
Activities
Terminal
Oct 26 16:56
siege@siege: ~/Desktop/assignment-3/source
siege@siege:~/Desktop/assignment-3/source$ strace sh hello-world.sh
execve("/usr/bin/sh", ["sh", "hello-world.sh"], 0xffffd7898d18 /* 45 vars */) = 0
brk(NULL) = 0xaaaaecc4d000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xffff99e64000
faccessat(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=62907, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 62907, PROT_READ, MAP_PRIVATE, 3, 0) = 0xffff99e1f000
close(3) = 0
openat(AT_FDCWD, "/lib/aarch64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\340u\2\0\0\0\0"..., 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=1641496, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 1810024, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xffff99c65000
mmap(0xffff99c70000, 1744488, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xffff99c70000
munmap(0xffff99c65000, 45056) = 0
munmap(0xffff99e1a000, 20072) = 0
mprotect(0xffff99df9000, 61440, PROT_NONE) = 0
mmap(0xffff99e08000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x188000) = 0xffff99e08000
mmap(0xffff99e0e000, 48744, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0xffff99e0e000
close(3) = 0
set_tid_address(0xffff99e64f50) = 4769
set_robust_list(0xffff99e64f60, 24) = 0
rseq(0xffff99e65620, 0x20, 0, 0xd428bc00) = 0
mprotect(0xffff99e08000, 16384, PROT_READ) = 0
mprotect(0xaaaac660b000, 8192, PROT_READ) = 0
mprotect(0xffff99e69000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0xffff99e1f000, 62907) = 0
getuid() = 1000
getgid() = 1000
getpid() = 4769
rt_sigaction(SIGCHLD, {sa_handler=0xaaaac65f26e4, sa_mask=~[RTMIN RT_1], sa_flags=0}, NULL, 8) = 0
geteuid() = 1000
getrandom("\x07\xc0\xa4\xcf\xbb\x60\x8f\x1d", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0xaaaaecc4d000
brk(0xaaaaecc6e000) = 0xaaaaecc6e000
getppid() = 4766
newfstatat(AT_FDCWD, "/home/siege/Desktop/assignment-3/source", {st_mode=S_IFDIR|0700, st_size=4096, ...}, 0) = 0
newfstatat(AT_FDCWD, ".", {st_mode=S_IFDIR|0700, st_size=4096, ...}, 0) = 0
openat(AT_FDCWD, "hello-world.sh", O_RDONLY) = 3

Step 3: Record System Calls

System Calls	C	C++	Java	Python	Bash
execvc	1	1	1	1	1
open()/openat()	2	5	29	49	48
close()	2	5	9	49	47
read()	1	4	8	68	62
write()	1	1		1	1
mmap()	6	18	25	24	24
Any System Calls	32	69	149	648	624
strace output file	hello-world.c.out.summary	HelloWorld.summary	hello-world.c.out.summary	hello-world.py.summary	hello-world.sh.summary

Step 4: Conclusion

Why does one Hello World program run slower or faster than another? What conclusion can you draw on the performance of the five programs from the information filled in the table? Defend your answer including the assumptions that you made concerning the CPU time.

CPU time to execute Program on a zHz CPU is estimated by:

$$CPUTime = NumberOfInstructions \times AverageInstructionCostCPUCycles \times \frac{1}{z}$$

The biggest factor for Hello World programs run faster or slower is the **number of instructions** they execute. As we can see from `strace`, Python and Bash make more than 600 System Calls while C and C++ make not more than 100. Also, programs made by languages other than C and C++ also call a lot of other kinds of System Calls such as `rt_sigprocmask`, `rt_sigaction`, and `fcntl`.

The average time per System Call, which is `usecs/call`, is another factor for impacting performance for languages like Java and Bash. As `HelloWorld.summary` shows, the `usecs/call` for `futex` is 13343 in Java, which a way beyond other languages (only 48 in C++).